

Phase 5: Project Development Phase

AI Brand Voice & Marketing Content Generator

1. Core Business Logic Implementation

The logical back-end of the application was engineered using modular Python functions. It structures specialized prompts and establishes few-shot learning boundaries to guide the Gemini LLM. Below are the finalized functions representing the logical heart of the project:

```
def learn_brand_voice(sample_texts):
    prompt = (
        "You are an AI marketing writer. Analyze these samples to deeply learn "
        "their tone, vocabulary, emotional style, and sentence rhythm.\n\n"
    )
    for i, text in enumerate(sample_texts, start=1):
        prompt += f"Example {i}: {text.strip()}\n"
    prompt += (
        "\nFrom these examples, infer the following:\n"
        "- Writing tone (e.g., formal, playful, confident)\n"
        "- Emotional attitude (e.g., inspiring, friendly, authoritative)\n"
        "- Vocabulary patterns (e.g., simple, descriptive)\n"
        "- Sentence rhythm and perspective\n"
        "\nUse this style in all future content generations. Do not output analysis."
    )
    return prompt

def generate_brand_content(brand_prompt, brand_name, purpose, content_type):
    topic = f"{brand_name} - {purpose}"
    templates = {
        "Social Media Post": "Include hooks, emoji usage, and clear CTA...",
        "Email": "With subject line, greeting, body, and sign-off...",
        "Tagline": "With concise, catchy options..."
    }
    content_template = templates.get(content_type, "Generate for " + content_type)
    final_prompt = f"{brand_prompt}\n\n{content_template}"
    try:
        response = model.generate_content(final_prompt)
        return response.text.strip()
    except Exception as e:
        return f"[Error] Failed to generate content: {str(e)}"
```

2. CSS Styling & Theme Integration

To elevate the visual styling above standard Streamlit layouts, we injected raw CSS styling. This defines custom gradients, shadow cards, and hover behaviors, providing a premium, native app feel:

```
.main-header {
    font-size: 3.5rem;
    background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
    text-align: center;
    margin-bottom: 0.5rem;
    font-weight: 800;
}

.feature-card {
    background: linear-gradient(135deg, #f8fafc 0%, #e2e8f0 100%);
    border-radius: 15px;
    padding: 1.5rem;
    border-left: 5px solid #2563eb;
    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.05);
    transition: all 0.3s ease;
```

}